ALGEBRA 1 · UNIT 8: EXPONENTIAL FUNCTIONS

Going Viral

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

A video of the robotics team is spreading online. Students model the views with exponential functions, compare them with a linear poster campaign, model the decay after the trend peaks, and design a viral challenge of their own.

Students compare repeated addition with repeated multiplication. The poster campaign starts ahead, but the exponential model passes it on day 6, which is a memorable moment. Decay after the peak shows factors less than 1. Exponent rules appear naturally when comparing views on different days.

Standards Addressed

- F-LE.1 Distinguish between linear and exponential situations
- F-LE.2 Construct exponential functions from a table or description
- F-LE.3 Observe that exponential growth eventually exceeds linear growth
- A-SSE.3c Use properties of exponents to transform expressions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: Would you rather have 400 views per day or views that triple each day starting from 5? Vote, then Part 1.
Day 2	Part 2: growth and decay rates and exponent rules.
Day 3	Part 3: design and pitch a viral challenge.

Materials

- Student handout
- Calculators
- Graph grid (provided)
- Optional: a short clip about how videos trend

How the Three Versions Differ

Level	What changes
CP2	Tables and patterns lead to $y = a(b)^x$; one decay model; exponent rules through 'how many times more views'.
CP1	Percent growth and decay, finding when exponential passes linear, and simplifying expressions with exponent rules.
Honors	Finds a and b from two data points, estimates doubling and half-life times, interprets negative exponents, and designs a model with constraints.

Watch For (Common Misconceptions)

- Writing $5 \cdot 3d$ or 15^d instead of $5(3)^d$.
- Using 0.4 instead of 0.6 for a 40% decrease.
- Assuming exponential is always bigger than linear (it starts smaller here).

Extension Ideas

- Research real view counts for a trending video and test the model.
- Use a logistic idea: why can't views triple forever?
- Compare compound interest to the viral model.

Answer Key – CP2

Part 1: Shares vs. Posters

#	Task	Answer
1	Complete the table.	Shares: 5, 15, 45, 135, 405, 1,215, 3,645. Posters: 0, 400, 800, 1,200, 1,600, 2,000, 2,400.
2	What do you multiply by each day for shares? What do you add each day for posters?	Shares: multiply by 3. Posters: add 400.
3	Write an equation for the shares model and one for the poster campaign.	$V = 5(3)^d$ and $V = 400d$
4	On which day do shares first have more views than posters?	Day 6 (3,645 > 2,400). On day 5, 1,215 < 2,000.
5	Plot both models on the grid for days 0 to 6.	Exponential curve rising sharply after day 4; straight line for posters.

Part 2: Growth Rules & the Fade

#	Task	Answer
6	On day d, the shares model gives $5 \cdot 3^d$ views. Two days later it gives $5 \cdot 3^{d+2}$. Use the rule $3^{d+2} = 3^d \cdot 3^2$ to explain how many times more views...	$3^{d+2} = 3^d \cdot 9$, so there are 9 times as many views.
7	Simplify $3^4 \cdot 3^2$ and $3^6 \div 3^2$. What do these mean for the video?	$3^6 = 729$ (4 days then 2 more days of tripling is 6 days). $3^4 = 81$ (views on day 6 are 81 times views on day 2).
8	After the video peaks at 10,000 daily views, daily views drop by 40% each day. What do you multiply by each day? Write an equation.	Multiply by 0.6. $y = 10{,}000(0.6)^d$
9	Find the daily views 3 days after the peak.	$10{,}000(0.6)^3 = 10{,}000(0.216) = 2{,}160$ views

Part 3: Design a Viral Challenge

#	Task	Answer
10	Choose how many people start it (a) and how many new people each person brings in (the multiplier b). Write your model.	Answers vary.
11	Make a table for days 0 to 6. Using the $5(3)^d$ model, on what day would a video pass 1,000,000 views? Use a calculator to extend the table.	Day 12: $5(3)^{11} = 885{,}735$ and $5(3)^{12} = 2{,}657{,}205$.
12	Explain why the challenge can't keep growing this way forever.	Answers vary (there are only so many people).

Answer Key – CP1

Part 1: Shares vs. Posters

#	Task	Answer
1	Write both models. Complete the table for days 0 to 6.	$V = 5(3)^d$; $V = 400d$. Shares: 5, 15, 45, 135, 405, 1,215, 3,645. Posters: 0, 400, ..., 2,400.
2	Graph both models and find the first day the shares model is ahead.	Day 6.
3	Explain the difference between adding 400 each day and multiplying by 3 each day, using the table.	Adding 400 gives equal steps; multiplying by 3 makes each step 3 times bigger than the last.
4	The club's account had 800 followers and grows 12% per week. Write a model and find the number of followers after 8 weeks.	$F = 800(1.12)^w$; $F(8) = 800(1.12)^8 \approx 1{,}981$ followers

Part 2: Exponent Rules & Decay

#	Task	Answer
5	Simplify $V(5) \div V(2)$ for $V(d) = 5(3)^d$. Interpret.	$5 \cdot 3^5 \div (5 \cdot 3^2) = 3^3 = 27$; day 5 has 27 times as many views as day 2.
6	Simplify $(5 \cdot 3^4)(3^2)$ and $(5 \cdot 3^8) \div (5 \cdot 3^8)$. What does the second result tell you?	$5 \cdot 3^6 = 3{,}645$ (day 6 views). The quotient is $3^0 = 1$, so any day divided by itself is 1.
7	After the peak (10,000 daily views), daily views decrease 40% per day. Write a model and find the first day with fewer than 1,000 daily views.	$y = 10{,}000(0.6)^d$. Day 4: 1,296; day 5: 778. Day 5.
8	A different account adds 150 followers per week starting at 800. In what week does the 12% growth account pass it?	Linear: 800 + 150w. Week 8: 1,981 vs. 2,000. Week 9: 2,218 vs. 2,150. Week 9.

Part 3: Design a Viral Challenge

#	Task	Answer
9	Write a growth model with a percent rate and a decay model for after the peak.	Answers vary.
10	Make a table and graph for both phases (at least 10 days in all).	Answers vary.
11	Compare your model to a linear plan of your choice. When does yours pass it?	Answers vary.

Answer Key – Honors

Part 1: Build the Model

#	Task	Answer
1	On day 2 the video had 180 views and on day 5 it had 4,860. Find a and b. Show your algebra.	$a \cdot b^2 = 180$ and $a \cdot b^5 = 4860 \rightarrow b^3 = 27 \rightarrow b = 3$; $a = 180 \div 9 = 20$. $V(d) = 20(3)^d$
2	Show that $V(d + k) \div V(d) = b^k$ for any d. What does this mean for the video?	$a \cdot b^{d+k} \div (a \cdot b^d) = b^k$. Any k-day period multiplies views by the same factor.
3	Find V(−1). What could it mean? Is it a reasonable prediction?	$V(-1) = 20 \cdot 3^{-1} = 20 \div 3 \approx 6.7$ views 'one day before day 0'. Not meaningful; the video did not exist yet.
4	Compare $V(d) = 20(3)^d$ with a linear campaign L(d) = 1,500d. On which day does V pass L, and when (if ever) does L lead?	Day 0: 20 vs. 0 (V ahead). L leads on days 1 to 5 (e.g., day 5: 4,860 vs. 7,500). Day 6: 14,580 vs. 9,000, so V passes L on day 6.

Part 2: Doubling & Half-Life

#	Task	Answer
5	Followers grow 12% per week from 800. Estimate the doubling time to the nearest week. Does the doubling time depend on the starting amount? Prove it.	$1.12^6 \approx 1.97$ and $1.12^7 \approx 2.21$, so about 6 to 7 weeks (≈ 6.1). $a(1.12)^t = 2a \rightarrow 1.12^t = 2$, so a cancels.
6	After the peak, views fall 40% per day. Find the half-life (to the nearest tenth of a day) by testing values.	$0.6^1 = 0.6$, $0.6^{1.4} \approx 0.49$, $0.6^{1.35} \approx 0.50 \rightarrow$ about 1.4 days.
7	Write the 12% weekly growth as a daily growth factor (to 4 decimal places). Explain your method using exponent rules.	Seven daily factors b must equal one weekly factor: $b^7 = 1.12$, so $b = \sqrt[7]{1.12} \approx 1.0163$ (power of a power: $(b^7)^w = b^{7w}$).
8	Simplify $(2x^3y^{-2})^3 \div (4x^5y^{-4})$ and explain how this skill helps you rewrite models.	$8x^9y^{-6} \div (4x^5y^{-4}) = 2x^4y^{-2} = 2x^4 \div y^2$

Part 3: Engineer a Viral Challenge

#	Task	Answer
9	Choose a and b so that $V(9) < 1{,}000{,}000 \leq V(10)$. Show the calculations.	Example: a = 50, b = 2.7: V(9) ≈ 381,000 and V(10) ≈ 1,029,000.
10	Add a decay phase after day 12 with a half-life between 2 and 3 days. Give the factor and show it meets the rule.	Example factor 0.75: $0.75^2 = 0.5625$, $0.75^3 \approx 0.42$, so the half-life is between 2 and 3 days.
11	Present both phases on one graph and explain the meaning of a and b in your models.	Answers vary.

Scoring note: Doubling time for 12% weekly growth is about 6.1 weeks; accept 6 weeks with reasoning. Accept half-life answers from 1.3 to 1.4 days.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original challenge uses a correct exponential model with a realistic growth factor. Goes beyond the minimum with a creative, realistic choice.	Original challenge uses a correct exponential model with a realistic growth factor.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Pitch includes the model, a table and graph, a linear vs. exponential comparison, and a decay plan. Polished, organized, and visually engaging.	Pitch includes the model, a table and graph, a linear vs. exponential comparison, and a decay plan. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.